

Gao Ying

Gender: Female Objective: Senior Backend Engineer Location: Hangzhou

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## Education

2010.09 - 2014.06 Nanjing University Data Science and Big Data Technology (B.S.)

Courses: Database Systems, Operating Systems, Machine Learning, Software Engineering, Design Patterns, Computer Networks

GPA 3.8/4.0, top 15% in major; merit scholarship

## Summary

Seven years of backend experience focused on payment/transaction systems, distributed architecture, high concurrency and production incident response.

## Work Experience

2017.07 - Present Bilibili Senior Backend Engineer

- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings, recognized by stakeholders and peers.
- Built service observability covering API latency, error rate, JVM GC, thread pools and consumer backlog with Prometheus and Grafana, reused across multiple business lines.
- Served 3350K requests per day while keeping core-path SLA above 99.9%.
- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices, and key metrics improved steadily.
- Designed a Kafka-based async transaction pipeline with idempotency keys, retry queues and dead-letter handling, sustaining 5000 QPS at peak, improving end-to-end delivery quality.

2014.07 - 2017.06 Huawei Backend Engineer

- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, with experience documented and shared internally.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 36%, and related standards were adopted by the team.
- Optimized a file-sort slow SQL on a hot API, added a composite index via gh-ost online DDL, reducing P99 from 900ms to 380ms, reused across multiple business lines.
- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 22%.

## Projects

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, and it became a reusable team best practice.
- Added circuit breaking to isolate faulty downstreams, and it became a reusable team best practice.
- Integrated OAuth2 and RBAC with multi-tenant isolation.
- Shipped a unified SDK to lower integration cost across teams, with experience documented and shared internally.

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Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Used sharding and read-write splitting to keep single-table rows within tens of millions, improving end-to-end delivery quality.
- Recorded an audit trace id for every payment state transition for compliance, recognized by stakeholders and peers.
- Applied idempotency and eventual consistency to avoid duplicate charges on retry.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction.

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Designed distributed locks and token-bucket rate limiting to prevent oversell, improving end-to-end delivery quality.
- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes, and key metrics improved steadily.
- Built circuit breaking and fallback for downstream failures, with experience documented and shared internally.
- Stayed stable at 5000 QPS in load tests with error rate below 0.1%.

#### Highlights

- Continuously optimized data analysis and metrics, cutting key-path latency from 900ms to 380ms
- Established metrics and reviews around JVM tuning and troubleshooting to keep improving delivery quality
- Built a reusable methodology and docs around requirement analysis and documentation, adopted by several teams
- Drove a focused effort on CI/CD pipelines, reducing related issues by about 36%
- Owned core modules that steadily served 3350K daily visits with 99.9%+ availability

#### Skills

Core skills: Microservices, Observability, Spring Boot, Kafka, SQL Optimization

Proficient in: MySQL, Java, Docker and containerization, Redis caching and distributed locks, High Concurrency

Familiar with: JVM, JVM tuning and troubleshooting, MySQL indexing and slow-query tuning, Redis, Kafka/RocketMQ messaging, Distributed transactions and consistency

#### Self Evaluation

- Curious about new tech, continuously learning LLM and cloud-native topics and applying them in practice
- Solid engineering fundamentals and strong troubleshooting skills, able to own a module end to end